Open Play Analysis

Gaming Habits and Mental Wellbeing amongst adults aged 18–40

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Introduction

Open Play (v1.2.6) is an open dataset built by researchers from the University of Oxford, Tilburg University and the Karolinska Institute. According to the official description, it links the gaming accounts of about 2,000 adults in the US and the UK (Steam, Xbox, Nintendo Switch and mobile) to psychological measures of mental health, motivation and wellbeing from 12 weeks of surveys, covering about 1.5M hours of play.[1]

Our submission contains:

- `open_play_analysis/`: the Bokeh server app, with `main.py` and one module per chart in `slides/`
- `Open_Play_Analysis.html`: the same presentation as a single HTML file
- `EDA/`: each member's notebook with the full data processing
- The processed CSV files read by the app, and this report

1. Who plays when? Do playtimes differ based on demographics?

We extracted the playtime as per our categories of demographics [Gender, Platforms used, Employment, Childcare] across all days, weekdays and weekends. Play is lowest around 5 a.m. and peaks around 8–9 p.m. on every platform and in every group: groups share the same daily rhythm, while the total amount depends on factors such as work and family.

Wellbeing Index

The surveys provide us a great opportunity to combine mental wellbeing measures such as PROMIS[2], Warwick-Edinburgh Measure[3] & Cantril Self-Anchoring scale[4] to create a composite Wellbeing index. It scales from 0 to 10, with higher numbers representing happiness/satisfaction in life.

2. Spikes, rings and runes: Playtime spikes and wellbeing

We perform time-series analysis by pltting the daily playtime against the wellbeing index and observe we spot two large spikes in playtime with corresponding fluctuations in wellbeing index around 2 points in time: March 2025 and June 2025. The exact dates of these spikes coincide with the release of some highly anticipated video games. Wellbeing index while spiking only during first playtime spike.

3. Loss of control, wellbeing index and playtime

We looked at the relationship between the wellbeing index and the feeling of losing control over gaming time. The more players report losing control, the more often they have low wellbeing, and the longer they spend gaming each day. Because loss of control, low wellbeing and long play go together, this part suggests that how people relate to their gaming may matter more than the hours alone.

4. Gender-wise game theme preferences

For participants with playtime registered of at least 1000 minutes, we extract information about their prefered games & the game metadata to analyze with their gender identity. Amongst the top 3 keywords, participants with Male or equivalent identities had higher occurances of the keyword "magic" whereas participants with Female or equivalent identities had higher occurances of "Male Protagonist".

5. Neurodivergence, wellbeing and playtime

We completed this graph as a bonus. Similar to above, we plot the wellbeing index with the playtime over two weeks with the Neuro-divergence diagnosis as a color map. While neuro-divergence status did not indicate any patterns, we are able to observe a relationship between the wellbeing and playtime.

A note on Accessibility

For Accessibility, we have kept text font at readable size with strong contrast of WCAG ratio of 14.27:1, which has the highest rating of AAA. We also checked our combinations on Colorsense Lab[5] for Colorblind safety and passed all tests. We have offered tooltips on relevant graphs for easier understanding.

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[1] Ballou, N., Földes, T. A., Vuorre, M., Hakman, T., Magnusson, K., & Przybylski, A. K. (2025). *Open Play: A longitudinal dataset of multi-platform video game digital trace data and psychological measures* [Preprint]. PsyArXiv. https://doi.org/10.31234/osf.io/nz96c_v1. Data: Open Play v1.2.6, Zenodo, https://doi.org/10.5281/zenodo.17536656

[2] Pilkonis, P. A., Choi, S. W., Reise, S. P., Stover, A. M., Riley, W. T., Cella, D., & PROMIS Cooperative Group. (2011). Item banks for measuring emotional distressfrom the patient-reported outcomes measurement information system (PROMIS®): Depression, anxiety,and anger. Assessment, 18(3), 263–283. https://doi.org/10.1177/1073191111411667

[3] Tennant, R., Hiller, L., Fishwick, R., Platt, S., Joseph, S., Weich, S., Parkinson, J., Secker, J., & Stewart-Brown, S. (2007). The warwick-edinburgh mental well-being scale (WEMWBS): Development and UK validation. Health and Quality of Life Outcomes, 5(1), 63. https://doi.org/10.1186/1477-7525-5-63

[4] Cantril, H. (1965). The pattern of human concerns. Rutgers University Press.

[5] https://colorsense.online/